Arista Access Node IND-ACC-65 Optical Transport Failure SOP (Minor Severity)

Node: IND-ACC-65 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_5984

Description

This Standard Operating Procedure (SOP) addresses the failure domain of Optical Transport Failure on the Arista Access Node IND-ACC-65, situated at Indiranagar Exchange, and primarily affecting the Access role. The root cause of such failures is attributed to the degradation of SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss. In an Arista network, the Optical Transport network is responsible for transmitting data between nodes over fiber optic cables. The SFP/XFP transceivers convert electrical signals into light signals that travel through the fiber, and vice versa. However, these transceivers can degrade over time due to environmental factors, such as temperature fluctuations, or physical damage, leading to signal attenuation and errors. Additionally, fiber cuts can occur due to accidental damage or aging of the fiber cables. The Rx/Tx power loss can also be caused by a combination of factors, including transceiver degradation, incorrect fiber connections, or faulty transceivers. In the event of an Optical Transport Failure, the Access Node will generate an alarm, and the network operator must perform diagnostic and resolution steps to isolate and mitigate the issue.

Pre-requisite Checks

- Ensure remote hands are dispatched to Indiranagar Exchange to perform any necessary physical checks or repairs.

- Verify the status of redundant links to ensure they are operational and can handle increased traffic loads.

- Review the network topology and configuration to identify any potential bottlenecks or single points of failure.

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces fiber status, Expected Output: Interface Name | Admin Status | Oper Status | Description, Description: This command displays the status of all fiber interfaces on the Access Node, including their administrative and operational states. The output will help identify any fiber interface errors or issues.
```

```
Command: show interfaces fiber detail, Expected Output: Interface Name | Admin Status | Oper Status | Description | Rx Power | Tx Power, Description: This command provides detailed information about each fiber interface, including their administrative and operational states, as well as their receive and transmit power levels. The output will help identify any issues with the SFP/XFP transceivers or fiber connections.
```

Resolution Steps

Step 1: Step: Replace faulty SFP/XFP transceiver, Command: replace interface fiber X/X/X with SFP-X/X/X, Description: If the diagnostic commands indicate that the SFP/XFP transceiver is faulty, replace it with a new one. Ensure the new transceiver is compatible with the Access Node and fiber connection.

Step 2: Step: Repair or replace fiber cable, Command: replace interface fiber X/X/X with fiber cable, Description: If the diagnostic commands indicate that the fiber cable is damaged or faulty, repair or replace it. Ensure the new cable is compatible with the Access Node and fiber connection.

Step 3: Step: Adjust Rx/Tx power levels, Command: set interface fiber X/X/X rx-power X and tx-power X, Description: If the diagnostic commands indicate that the Rx/Tx power levels are out of range, adjust them to the recommended values. Ensure the new power levels are compatible with the Access Node and fiber connection.

Post-Check Verification

```
Command: show interfaces fiber status, Expected Output: Interface Name | Admin Status | Oper Status | Description, Description: Run this command to verify that the fiber interface is operational and error-free.

Command: show interfaces fiber detail, Expected Output: Interface Name | Admin Status | Oper Status | Description | Rx Power | Tx Power, Description: Run this command to verify that the SFP/XFP transceiver and fiber connection are operating correctly and within the recommended power levels.
```

Roll-back Procedures

- Step: Restore original SFP/XFP transceiver, Command: replace interface fiber X/X/X with original SFP-X/X/X, Description: If the resolution steps do not resolve the issue, restore the original SFP/XFP transceiver to its previous state.

- Step: Restore original fiber cable, Command: replace interface fiber X/X/X with original fiber cable, Description: If the resolution steps do not resolve the issue, restore the original fiber cable to its previous state.

- Step: Reset Rx/Tx power levels, Command: set interface fiber X/X/X rx-power X and tx-power X to default, Description: If the resolution steps do not resolve the issue, reset the Rx/Tx power levels to their default values.